Correlation Dimension of Auto-regressive Large Language Models

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Abstract

Large language models (LLMs) have achieved impressive results in natural language generation but continue to exhibit unexplained behaviors—such as repetition and incoherence—even at low perplexity. This exposes the limitations of traditional evaluation metrics, which focus on local prediction accuracy while neglecting long-range structural complexity. We introduce the correlation dimension, a fractal-geometric measure of self-similarity, to quantify the implicit textual complexity perceived by language models. Correlation dimension robustly captures the hierarchical recurrence structure of text, bridging local and global properties. Experiments demonstrate that this measure reveals distinct pretraining phases, reflects context-dependent complexity, and reliably detects various forms of degeneration in generated text. The method is computationally efficient, broadly applicable to autoregressive models, and offers new insight into the generative dynamics of LLMs.

1 Introduction

Latest advances in large language models (LLMs) have demonstrated sophisticated capabilities, including mathematical reasoning and planning. These abilities suggest taht LLMs internally process information through complex, non-linear, and potentially hierarchical mechanisms, despite operating through simple, token-by-token predictions. Understanding precisely how LLMs achieve such macroscopic behaviors from microscopic predictive steps remains an important open question, essential for grasping the full potential and limitations of these models. Although previous research shows that minimizing next-token prediction loss—perplexity—is theoretically powerful [28], it still falls short in fully explaining unexpected model behaviors, such as hallucinations and repetitive, bland outputs, even when perplexity values are low. Thus, new tools for characterizing the complex behavior of LLMs are urgently needed.

Current evaluation approaches broadly fall into two categories. The first includes methods based on local textual properties, such as lexical or syntactic statistics (e.g., n-gram frequencies). These metrics are intuitive and interpretable but often fail to capture semantic ambiguities or deeper structural complexities. The second category consists of global metrics, such as mean perplexity or semantic similarity measures. While these methods provide comprehensive quantitative evaluations, they frequently lack interpretability and specific insights into local textual phenomena. This divide between local interpretability and global comprehensiveness reflects the broader challenge of bridging microscopic (token-level) and macroscopic (long-range, structural) perspectives of LLM behavior.

In this work, we propose *correlation dimension*, a measure drawn from fractal geometry and dynamical systems theory, to bridge this gap. Correlation dimension quantifies self-similarity—a fundamental characteristics of complex systems that exhibit invariant patterns across scales. Originally developed for analyzing deterministic chaotic systems [13], correlation dimension has since been successfully adapted to stochastic processes and real-world complex phenomena [35, 24]. Applied to language

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models, correlation dimension effectively the intrinsic complexity and recurrence structure of generated texts as perceived by the model. For instance, texts with randomly shuffled words appear highly complex and yield high dimensionality, whereas simple repetitive patterns exhibit low dimensionality.

We specifically propose computing correlation dimension using sequences of next-token log-probability vectors, which are readily available for any autoregressive language model. Recurrences are defined via the Euclidean distance between these vectors. Unlike perplexity, correlation dimension reflects deeper structural properties of the text generation process. For example, a model experiencing mode collapse might produce contextually irrelevant but statistically plausible text, exhibiting low perplexity yet high correlation dimension. Conversely, models lacking adequate memory to handle long-range dependencies might show low dimensionality despite high perplexity, clearly indicating their limited structural comprehension.

Correlation dimension serves as both a practical and theoretically grounded metric. It requires little computational overhead beyond perplexity calculations and runs at inference time, making it easy to be integrated into existing inference infrastructures such as `vllm`. At a theoretical level, it provides nuanced insights into model behavior, revealing the long-range complexity structures that perplexity alone cannot uncover. Practically, it offers a straightforward yet powerful method to evaluate model robustness, capturing subtle forms of degeneration such as repetition, incoherence, and blandness.

Throughout the paper, we illustrate how correlation dimension inherently connects local recurrence structures and global textual complexity (Section 3), provides insights into intrinsic properties of texts and model behavior (Section 4), and effectively detects degeneration issues in long-text generation (Section 5). Overall, our approach naturally integrates interpretability and comprehensiveness, offering a robust metric grounded in the fundamental properties of complex dynamical systems.

2 Related Works

2.1 Statistical Self-Similar Phenomenon in Language

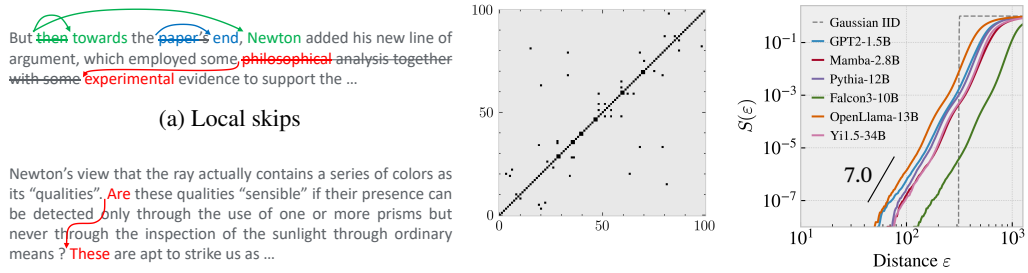
Self-similarity is a fundamental property observed in various complex systems, characterized by invariant patterns across multiple scales. A special class of self-similarity, known as scale-free properties, refers specifically to patterns that are consistent when viewed at different scales [42]. Unlike precise scale-free structures defined mathematically (e.g., the Koch snowflake), *statistical self-similarity* denotes approximate scale invariance identified through statistical analysis of empirical data [39].

Statistical self-similarities has been widely documented in natural language, manifesting through phenomena such as Zipf's law [48, 29, 31] for word frequencies, Herdan-Heap's law [20, 17] for vocabulary growth, and long-range correlations [10, 9, 2, 40] for word occurrence across text spans.

Recent approaches have explored statistical self-similarity at the abstract semantic level, extending beyond traditional lexical-based metrics (e.g., word frequency). For instance, [6, 7] utilized latent semantic analysis (LSA) embeddings of text paragraphs [5] and discovered geometric self-similarity (fractal structures) within semantic spaces, estimating fratal dimensions approximately between 8 and 20. More recently, consistent semantic self-similarity across multiple languages was observed using pretrained language models [8]. However, the linguistic significance of the self-similarity and the explicit role of language models in capturing such phenomena is unclear. This paper proposes a new method utilizing language models that naturally addresses these gaps, thereby providing a new analytical tool for characterizing large language models.

Characterizing LLMs via Self-Similarity Recent studies have evaluated language models via statistical self-similarity. [37] compared generated texts from language models to natural texts across five scale-free properties, revealing significant deviations in scaling exponents. Building on this, [30] introduced quantitative statistical tests to measure such deviations systematically and examined how different sampling strategies (e.g., nucleus sampling vs. beam search) affect self-similarity.

Notably, [1] recently measured the fractal dimension and Hurst exponent of cumulative log-perplexity series, demonstrating that these scaling exponents correlate with downstream performance of LLMs. However, their approach primarily captures long-range dependencies in the model's predictive error sequences. In contrast, our method (detailed in Section 3) examines the intrinsic generative recurrence



(a) Local skips (b) Long-range skips (whole sentences) (c) Recurrence plot (d) Correlation integral curves

Figure 1: Experimental results on the "Newton's Philosophy" article of the Stanford Encyclopedia of Philosophy. (a-b) Examples of textual skips as predicted by the Pythia-12B model, illustrating both local and long-range recurrences in the text. (c) Segment of the recurrence plot for the log-probability vectors generated by the model; black dots indicate pairs of points within a specified distance threshold. (d) Correlation integral curves for six pre-trained language models (solid lines), compared to i.i.d. Gaussian noise in $\mathbb{R}^{50000}$ (dashed lines).

structure inherent in language itself, uncovering fundamentally different insights about the model behavior and text complexity.

2.2 Complexity Metrics of Sequential Data

3 Recurrence Structure and Correlation Dimension of Language

Natural language exhibits self-similarity across various linguistic scales, ranging from morphological, lexical, and syntactic to semantic structures and long-range dependencies. Unlike mathematically precise fractals such as the Koch snowflake, real-world self-similarity is typically approximate and statistical, manifesting through recurrent statistical patterns rather than exact repetitions.

The *correlation dimension* provides a quantitative framework to measure such statistical self-similarity in sequences by analyzing their *recurrence* structure. A recurrence is defind as an event where the trajectory of a system approximately revisits a previous state within a predefined distance threshold ε. As ε increases, more recurrences naturally emerge. Self-similar systems typically follow a power-law relationship betwen the recurrence frequency $S(\varepsilon)$ and the distance threshold ε, formally defined as follows:

Definition 1 (Correlation dimension [13])**.** *Given an infinite sequence $\{x_t\}_{t=1}^{\infty}$ in a metric space (e.g., $\mathbb{R}^D$), its correlation dimension d is the exponent characterizing the scaling behavior of the correlation integral $S(\varepsilon)$:*

$$S(\varepsilon) \quad \propto \quad \varepsilon^d \quad \textit{as} \quad \varepsilon \to 0, \tag{1}$$

where the correlation integral $S(\varepsilon)$ is defined as the frequency of point pairs separated by distances less than ε:

$$S(\varepsilon) = \lim_{t\to\infty} \frac{2}{t(t-1)} \sum_{1\le i<j\le t} 1\{\|x_i - x_j\| < \varepsilon\}. \tag{2}$$

Here, $\|\cdot\|$ denotes the Euclidean norm, and $1\{\cdot\}$ is the indicator function that equals 1 if the condition is true and 0 otherwise.

Correlation dimension was originally developed to characterize deterministic attractors (e.g., Hénon maps [19]) but subsequently generalized for analyzing stochastic processes such as fractional Brownian motion [11] and complex networks [24].

Correlation Dimension Applied to Language. Representing text as a sequence of numerical vectors allow the identification and measurement of recurrences and subseqeuntly their correlation dimension. However, selecting a meaningful, stable, and interpretable representation for natural language is nontrivial.

We propose utilizing the Euclidean distance betwen *logarithmic* next-token probability vectors derived from language models. Specifically, the log-probability vector at time t, denoted x_t, is calculated as:

$$x_t(\omega) = \log \mathrm{P}_\theta(\omega_t = \omega | \omega_{t-c}, \cdots, \omega_{t-1}) \quad \forall \omega \in \Omega, \tag{3}$$

where ω_t represents the token at position t, P_θ represents the model-predicted probability (θ specifies the model), and Ω is the vocabulary set. Unless otherwise specified, the context length c is set to 512 tokens; its impact is further examined in Section 4.2.

Textual Skips as Recurrences. Recurrences in next-token log-probability vectors can be interpreted as *textual skips*. If two states x_t and x_s ($s < t$) are close, the text segment $[s, t)$ could theoretically be omitted without significantly altering subsequent text generation. Skips occur at multiple scales, as illustrated in Figure 1(a-b): from local (single-word skips) to global (sentence-level skips). Smaller distance thresholds (ε) identify local skips, while larger thresholds detect longer-range skips. The hierarchy aligns naturally with Chomsky's generative grammar [4], where such skips may correspond to omitted subtrees within the hierarchical structure of language.

Figure 1(c) provides an example recurrence plot based on log-probability vectors from the text *Newton's Philosophy* [23], clearly visualizing recurrences at a certain threshold. The correlation integrals $S(\varepsilon)$, illustrated in Figure 1(d), demonstrate near-linear scaling for multiple pre-trained language models (GPT2 [34], Pythia [3], Falcon3 [41], OpenLLaMA [12], Yi1.5 [44], and Mamba [16]), yielding a consistent correlation dimension around 7. In contrast, Gaussian random vectors in equivalent dimensional spaces exhibit distinct different scaling behavior, underscoring language-specific recurrence structures.

3.1 Sufficiency of Next-Token Log-probabilities

The next-token log-probability vectors represent only partial information of a language model's full state, defined theoretically as the distribution over all future token sequences. Thus, one might question the sufficiency of next-token probabilities alone for characterizing long-range text generation.

Time-delayed embeddings, motivated by Takens' embedding theorem [38] and its stochastic extensions [35, 36], are common methods for reconstructing full states from partial observations. Such embeddings concatenate multiple sequential log-probability vectors, $y_t = [x_t; x_{t+1}; \ldots; x_{t+k}]$, potentially capturing higher-order dependencies.

However, empirical observations reveal negligible differences in the correlation dimension when comparing simple next-token log-probability vectors ($k = 1$) to embeddings with higher orders ($k > 1$), after accounting for the noise inevitably introduced in the delayed embeddings. This somewhat surprising result suggests that single-step log-probability vectors inherently encode significant long-term structural information about language evolution. This phenomenon mirrors findings in knowledge distillation [21], where single-step probability distributions effectively summarize model knowledge. Additional details and analysis are provided in Appendix C.

3.2 Empirical Convergence of Correlation Dimension

Considering an ideal "perfect" langauge model capable of precisely predicting next-token distributions, its correlation dimension would reflect intrinsic textual complexity. However, slight imperfections in predictions significantly affect correlation dimension measurements, particularly due to rare outcomes influencing the measure disproportionately compared to typical loss calculations. Precisely, let $l(\theta) = \mathbb{E}_{\omega_{\leq t} \sim \mathbb{P}}\big[\log \mathrm{P}_\theta(\omega_t | \omega_{\leq t-1})\big]$ be the cross-entropy loss function of a language model parameterized by θ, where $\mathbb{P}$ and P_θ represent the empirical distribution and the model-predicted distribution, respectively, and $\omega_{\leq t} = [\omega_1, \cdots, \omega_t]$ represents the sequence of tokens until time t. Then, we can see that the contribution of rare outcomes ω to the total loss is very small:

$$\frac{\partial\, l(\theta)}{\partial\, \log \mathrm{P}_\theta(\omega | \omega_{\leq t-1})} = \mathbb{P}(\omega | \omega_{\leq t-1}) \qquad \forall \omega \in \Omega. \tag{4}$$

In other words, the log-probability vectors at rare words ω may vary significantly, without affecting the loss function, but this variation will be reflected in the correlation dimension regardless of the rareness of the word.

Nonetheless, empirical experiments indicate convergence in correlation dimensions of well-trained language models to a narrow range as perplexity decreases. Figure 2 illustrates correlation dimension for various pre-trained LLMs across the SEP dataset which is summarized in Appendix B. As the perplexity decreases, correlation dimensions stablize around a consistent value near 6.5. This

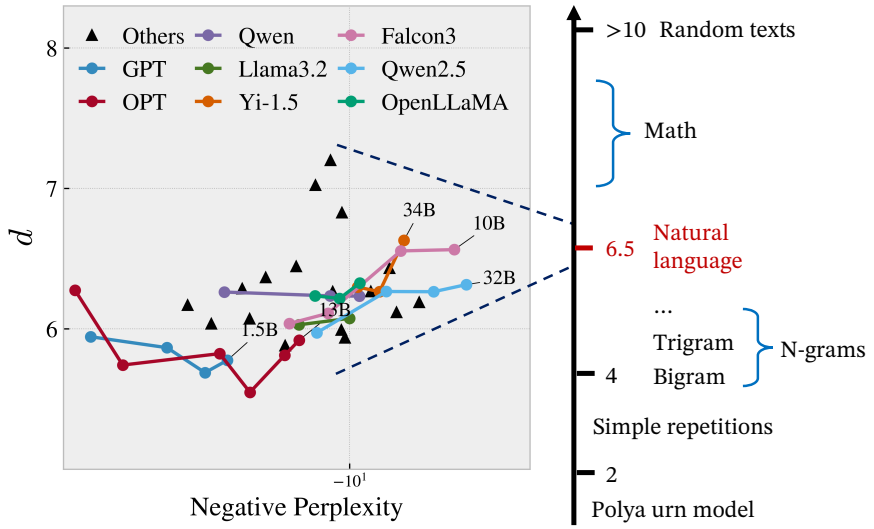


Figure 2: Left: Mean correlation dimension of the SEP dataset measured with various pre-trained LLMs. Right: A conjectured spectrum of correlation dimension for different types of texts / language models.

empirical convergence underscores correlation dimension's reliability as a stable, interpretable metric for evaluating the structural complexity of natural language and the performance of language model.

The right half of Figure 2 presents the spectrum of correlation dimension values across various types of texts and statistical processes. Randomly shuffled texts exhibit high correlation dimensions, typically above ten, whereas self-reinforcing processes such as the Polya urn model [27] display much lower values, below two. n-gram processes with small n also yield lower correlation dimensions compared to natural language, while mathematical texts occupy the higher end of the spectrum. The latter observation is based on the MATH dataset [18] analyzed with a math-finetuned LLM; however, this finding remains provisional, as the characteristics of "typical" mathematical texts are not well defined—particularly given the rapid evolution of math-enhanced LLMs.

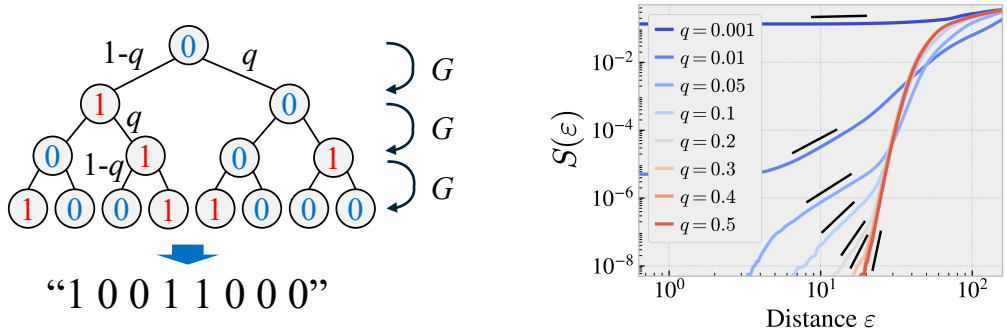
4 Characterizing Language Models Using Correlation Dimension

Estimating the correlation dimension of a text using language models raises critical questions regarding the reliability and interpretability of this measure when the underlying model is imperfect or incompletely trained. Ideally, with a perfect language model, the correlation dimension would represent the intrinsic complexity of the text itself. However, real-world language models inevitably contain imperfections, prompting us to investigate whether the correlation dimension remains meaningful under such conditions.

In this section, we demonstrate that the correaltion dimension is a robust measure of the *perceived complexity* of texts from the perspective of language models, even when these models are imperfect or incompletely trained. Specifically, we explore three key aspects: (1) how the correlation dimension reflects the inherent complexity of a text as determined by its underlying generative structure (Section 4.1); (2) how the correlation dimension is influenced by constraints on the contextual information available to the model (Section 4.2), and (3) how distinct stages emerge in the correlation dimension throughout the training process of language models, a phenomenon not observed in conventional metrics such as perplexity (Section 4.3).

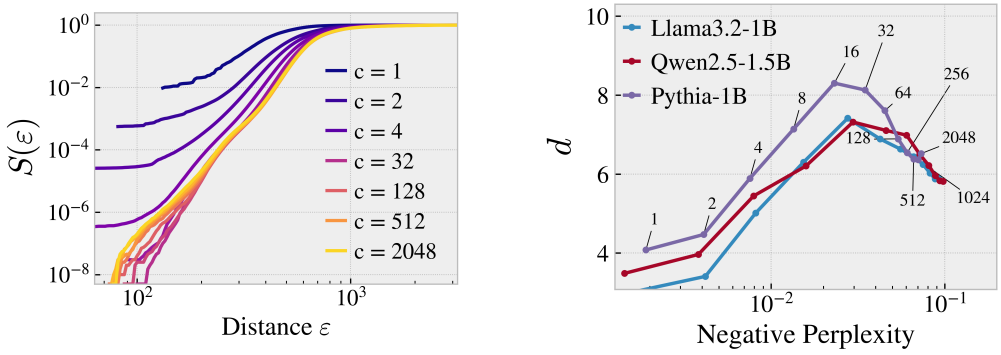
4.1 Correlation Dimension and Textual Complexity

To analyze how correlation dimension captures inherent textual complexity, we generate sequences using the Lin-Tegmark grammar [26], a probabilistic context-free grammar defined over a binary alphabet $\{0, 1\}$. The grammar's complexity is parameterized by a single Bernoulli parameter $q \in$



(a) (b)

Figure 3: (a) Illustration of the Lin-Tegmark grammar generation process governed by the transition matrix G (Eq. (5)), parameterized by q. (b) Correlation integral curves for texts generated with different q values, measured using the OpenLLaMA-13B model.



(a) (b)

Figure 4: (a) Correlation integral curves on an article from the SEP dataset, measured by Pythia-1B model at different context lengths. (b) Mean correlation dimension on the SEP dataset (vertical axis), measured using three models at different context lengths, with respect to the negative perplexity (horizontal axis).

$[0, 1]$ (Figure 3(a)), controlling the mutual information decay between tokens:

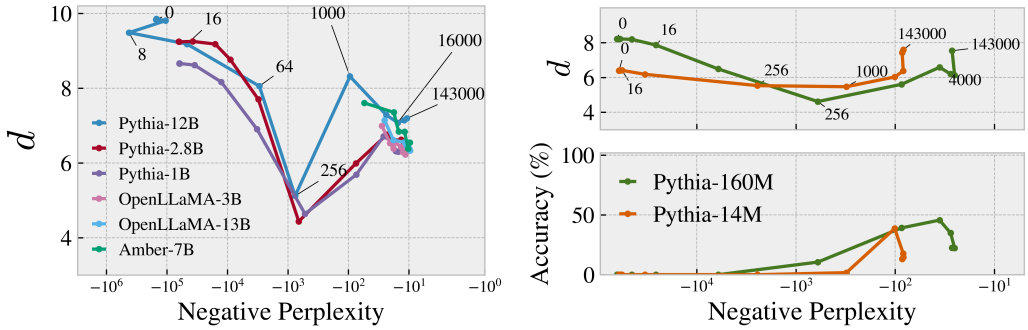
$$G = \begin{pmatrix} q & 1-q \\ 1-q & q \end{pmatrix}, \tag{5}$$

where G_{ij} denotes the probability of generating a child node valued j given a parent node valued i. The resultant text complexity varies from highly predictable sequences (as q approaches 0 or 1) to highly unpredictable sequences (as q approaches 0.5).

Figure 3(b) illustrates the correlation integral curves computed by the OpenLLaMA-13B model for different values of q. As q increases towards 0.5, the correlation dimension increases markedly from near zero to values above ten, demonstrating its sensitivity to textual complexity. Despite the nonlinear curves due to the model not being explicitly trained on this grammar, the correlation dimension consistently captures the inherent complexity defined by the grammar.

4.2 Effect of Contextual Constraints on Correlation Dimension

We next explore how limiting contextual access affects the correlation dimension measured by pretrained language models. Varying the context length parameter c in Eq. (3) imposes restrictions on the model's available context, directly influencing its complexity perception. A context length of $c = 1$ reduces the model effectively to a bigram approximation, while longer contexts progressively enable deeper linguistic comprehension.



(a) (b)

Figure 5: (a) Evolution of the mean correlation dimension for various language models on the SEP dataset during pre-training. (b) Notable case of two smaller models (Pythia-14M and 160M) that show a marked increase in correlation dimension (top) near the end of training, coinciding with a significant drop in in-context learning accuracy (bottom).

Figure 4(a) shows correlation integral curves for the Pythia-1B model measured on the SEP dataset at varying context lengths. Figure 4(b) depicts average correlation dimensions against perplexity for multiple models (Pythia-1B [3], Qwen2.5-1.5B [43], and Llama3.2-1B [14]). Notably, we observe a two-stage pattern: initial correlation increases sharply from approximately 3 to about 8 as context length extends to 32 tokens, followed by a gradual reduction to around 6.5 at longer contexts.

This pattern indicates that initial increases in context length enhance the model's perception of complexity, revealing more contextual variation. Subsequently, the model identifies redundant contextual variations, compressing perceived complexity and converging to a stable dimension around 6.5. This trend consistently appears across different models, validating the correlation dimension as a reliable metric of the complexity perceived by language models.

4.3 Three-Stage Evoluation in LLM Pre-Training

Pre-training language models typically focus on monotonically reducing perplexity, implying a linear improvement. However, we find that correlation dimension reveals a distinct, nonlinear evolution consisting of three stages during training.

We monitored correlation dimension across multiple checkpoints of various models, including the Pythia family (12B, 2.8B, 1B, 160M, 14M), OpenLLaMA (3B, 13B), and Amber-7B (Figure 5). The evolution distinctly unfolds in three stages: (1) an initial rapid drop in correlation dimension due to learning short-range (bigram-level) structures, (2) a subsequent increase as models begin capturing longer-range dependencies, and (3) a final gradual decline indicating improved generalization via context compression. This three-stage evolution is clearly observed in the Pythia family, for which the early-stage checkpoints are available; for the other models, the evolution is verified in the last stage. Notably, these shifts occur even though perplexity continuously decreases.

Interestingly, the third stage, indicative of improved generalization, does not universally occur. Smaller models, like Pythia-14M and Pythia-160M, instead exhibit a sudden increase in correlation dimension at later stages, correlating with degraded performance in contextual learning tasks. Near the end of the training process, the correlation dimensions of these models exhibit a sudden rise to around 8. In the lower half, we show the accuracy of the two models in a simple in-context learning task, where the model is asked to repeat a sequence of symbols. A clear correlation is seen between the accuracy and the increase in correlation dimension, suggesting the loss of generalization ability of the model. This observation underscores the potential of correlation dimension to indicate generalization failures and stability in language models, insights inaccessible through perplexity alone.

5 Correlation Dimension for Quantifying Degeneration in Text Generation

A critical challenge in language model generation is maintaining coherence, diversity, and relevance throughout extended text sequences. A well-known phenomenon, termed *degeneration*, describes scenarios where texts become repetitive, incoherent, or bland [22]. While explicit repetition is easily detectable, subtler forms of degeneration such as incoherence or blandness lack clear definitions or reliable quantification methods.

Table 1: Schematic summary of various LLM evaluation methods (rows) and their ability to detect different types of degeneration (columns). Triangles indicate potential usefulness that has not been empirically validated.

Metric	repetition	incoherent	bland
Local generation probability			
Perplexity	△	×	△
Cond. Entropy	△	×	△
Word or N-gram statistics			
Zipf Coefficient [48]	✓	△	×
Heap Coefficient [17]	✓	△	×
Rep-N [22]	✓	×	×
Distinct-N [25]	✓	×	×
Self-BLEU [47]	✓	×	△
Semantics			
BERTScore [46]	×	✓	△
MAUVE [32]	×	✓	△
CorrDim (ours)	✓	✓	✓

In this section, we propose using correlation dimension as a unified metric to quantify degeneration. Conceptually, degeneration is viewed as a sudden collapse from a higher-dimensional trajectory in the model's state space into a lower-dimensional attractor. Such collapses generally are irreversible, mirroring the *boundary crisis* phenomenon in chaotic dynamical systems [15]. Table 1 compares several puplar evaluation methods (rows) against specific degeneration types (columns), highlighting correlation dimension's unique capability to detect all considered forms: repetition, incoherence, and blandness.

5.1 Detecting Degeneration Beyond Repetition

To empirically evaluate our approach, we created a controlled dataset comprising responses to twenty generic questions. Each question had ten normal responses and intentionally degenerate texts—repetitive, incoherent, or bland—generated by the GPT-4o model (details in Appendix D). We computed correlation dimensions using the Falcon3-10B model [41]. Because all the texts are short and their lengths are within the context length limit of the model, we set c to the maximum context length of the model (See Section A).

The results, shown in Table 2, indicate significantly lower correlation dimensions for degenerate texts compared to normal responses, confirmed by Wilcoxon signed-rank tests (all p-values < 0.01). These findings validate the correlation dimension's sensitivity to different degeneration modes, including those traditionally difficult to quantify, such as incoherence and blandness.

Table 2: Correlation dimension or perplexity of degenerate texts compared to normal texts, measured using the Falcon3-10B model, with p-values from Wilcoxon signed-rank test in the parentheses.

Group	mean CorrDim	mean perplexity
Normal	5.04	10.79
Repetitive	3.80 (9.5E-7)	1.25
Incoherent	3.96 (2.9E-6)	13.24
Bland	4.51 (1.1E-3)	4.24

For comparison, the table also reports the mean perplexity (right-most column) of each group, as measured by the Falcon3-10B model. While perplexity can distinguish certain types of degeneration—show much lower values for repetitive texts and higher values for incoherent texts compared to normal response—the direction of change is inconsistent across degeneration types. This indicates perplexity not being a intrinsic measure of degeneration.

5.2 Stress-Testing Language Models with Random Texts

To access model susceptibility to degeneration, we designed a stress-test involving random texts—specifically, a list of randomly generated English names separated by commas, totaling approximately 1024 tokens. Models were tasked with completing the sequence from varying input lengths (n), generating $1024 - n$ tokens. Random sequences typically have high initial correlation dimension due to unpredictability; thus models prone to degeneration exhibit abrupt drops in dimension as repetition or incoherence emerges.

Figure 6 illustrates correlation dimension trends for three models as the input length n increases (and the output length $1024 - n$ decreases). Robust models like Yi1.5-34B consistently increasd their correlation dimension, whereas weaker models such as Qwen2-7B-Instruct displayed marked dimension drops, indicating degeneration.

Further quantification is presented in Table 3, which reports mean correlation dimensions at input length $n = 512$ alongside scores from the HelloEval text-completion benchmark [33]. A high correlation (Spearman's $\rho = 0.952$) is evident, underscoring correlation dimension's validity as an intrinsic measure of model robustness in generating long, coherent text without relying on complex evaluation tasks.

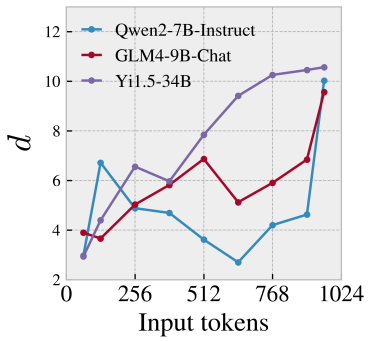


Figure 6: Correlation dimension of generated texts with respect to the length of input texts.

Table 3: Mean correlation dimensions of generated texts for each model, compared with the HelloEval text completion scores.

Model	HelloEval	CorrDim
Qwen2-7B-Instruct	5.12	3.54
Llama3.1-8B	-5.61	4.15
InternLM2.5-7B	6.39	4.61
Mistral-7B-v0.2	13.05	5.23
GLM4-9B-Chat	12.32	6.45
LongWriter-GLM4-9B	17.67	8.01
InternLM2.5-20B	36.68	8.89
Yi1.5-34B	44.73	8.89
Spearman's ρ	-	0.952

6 Practical Usage of Correlation Dimension

Lower CorrDim at a similar perplexity often indicates better generalization.

Detection of Explicit Memorization from inconsistency between models.

Sliding window triggers redo generation

7 Limitations

We acknowledge several limitations in our work. First, our approach requires accessing a language model's full output probabilities, which are not available typically for models available via APIs, such as GPT-4. This limits the evaluation of closed-source models. Second, it is unclear to which extent advanced quantization techniques would affect the correlation dimension. We conducted our experiments based on the FP16 precision at model inference and FP64 at distance calculation.

8 Conclusion

In this paper, we introduced correlation dimension as a robust and insightfu metric for analyzing the complexity and structural characteristics of language generation processes in large language models (LLMs). Unlike conventional evaluation metrics that focus primarily on prediction accuracy, the correlation dimension provides a quantitative measure of the perceived textual complexity from the perspective of the language model itself.

Our results show that correlation dimension captures inherent text complexity and reveals multi-stage learning behaviors during LLM training—patterns that remain hidden to perplexity-based evaluation. It highlights stages such as short-range dependency acquisition, long-range contextual structure emergence, and generalization or degeneration. Additionally, we validated correlation dimension as a comprehensive detector of various forms of degeneration—including repetition, incoherence, and blandness—in model-generated texts. Our empirical results confirmed its superior sensitivity compared to existing evaluation methods. Moreover, stress-testing experiments demonstrated correlation dimension's effectiveness in evaluating a model's robustness to degeneration in long text generation tasks.

Overall, the correlation dimension emerges as a versatile and theoretically grounded tool that complements traditional metrics, offering deeper insights into language model behavior, structural complexity, and generalization capability. Future research directions include further theoretical exploration of correlation dimension properties and practical applications in model evaluation and optimization.

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A Details for Estimating Correlation Dimension

In practice, we can only observe a finite sequence of points (i.e., finite t); in addition, there is no guarantee that the correlation sum $S(\varepsilon)$ is a perfect power-law function of ε. Nevertheless, the empirical correlation dimension still provides meaningful descriptions of the complexity of the system, as we will shown in the following sections. The correlation dimension d can be estimated by the slope of the log-log plot of $S(\varepsilon)$ versus ε, i.e., $\log S(\varepsilon) \propto d \log \varepsilon$. To provide accurate estimates which are consistent across sequences of different lengths t, we limit the range of $S(\varepsilon)$ to $[\frac{2}{t(t-1)} \times 10, \frac{1}{t} \times 1]$ and estimate the slope using an ordinary least-square fit. This range covers roughly the lower half of the curve while avoiding the tail that is often dominated by noise.

We adopt slightly different settings in estimating the correlation dimension for long and short sequences. For long sequences, such as those in the SEP dataset, which are beyond the context limit of typical LLMs, we used a moving-window approach to estimate the next-token prediction probability with a exact predefined context length (e.g., 512 tokens).

For short sequences, this approach would lead to discarding a large portion of the sequence. Therefore, we did not restrict the context length for short sequences. On the other hand, such short sequences are often within the context limit of the model, and the model can access all tokens in the sequence. In this case, we input the entire sequence all at once and calculate the correlation dimension using the output probabilities at all time steps. Therefore, the log-probability vectors are estimated with an increasing context length. This applies to the experiments in Section 5.

All the experiments throughout the paper were conducted on a single NVIDIA H100 GPU with 92GB memory. The amount of GPU memory required is approximately twice the parameter size of the model, as we used FP16 precision for model inference.

B The SEP Dataset and Additional Results

We constructed the Stanford Encyclopedia of Philosophy (SEP) dataset by using the 60 longest articles on the SEP website [45]. The identifiers of the articles are listed in Table 4 in the left-most column. For each article, we removed the prologue and catalog. All the 60 articles were longer than 20,000 words. In measuring the correlation dimension, we truncated each article to the first 20,000 tokens. This may introduce a slight difference between models that use different tokenizers, but the difference was small for most models we examined.

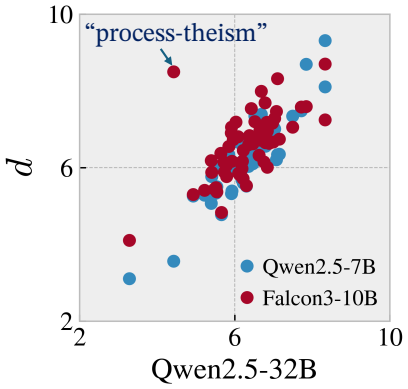

Figure 7: Comparison of correlation dimensions of the SEP articles between three models.

Results for three representative models, Qwen2.5-32B, Qwen2.5-7B, and Falcon3-10B, are shown in Table 4. The second to fourth columns show the correlation dimensions of the articles, and the right-most column shows the perplexities for the Qwen2.5-32B model. The correlation dimensions among articles showed good consistency across different models, especially within the same family (Qwen2.5-32B), as shown in Figure 7.

The largest difference in correlation dimension was observed for the article "process-theism", as indicated in Figure 7 and in Table 4 in gray. which contained long lists of names. The Qwen2.5

models regarded the list of names as a flattened structure and estimated a low correlation dimension, while the Falcon3-10B model inferred a complex structure behind the list and estimated a high correlation dimension. This difference reflects an interesting aspect of the models in terms of their "perception" of text structure. The difference in correlation dimension could be due to the different training data and the fine-tuning process of the models.

C Consistency with Time-Delayed Embeddings

A potential concern with the correlation dimension estimation method in Section 3 is that we used only the probabilistic information of the next tokens, which is not a complete representation of the model's state. A state of a dynamical system is defined as a point in the phase space, which contains all the information that governs the future evolution of the system.

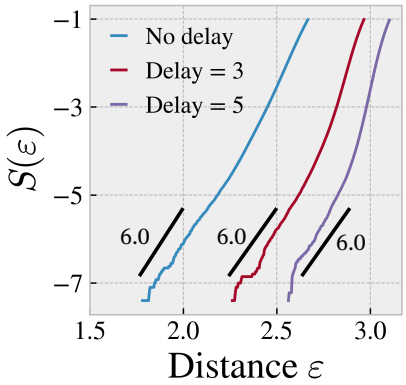

Figure 8: Correlation integral curves for time-delayed embeddings with different delays (3 or 5), compared with that for the original sequence.

A common method to reconstruct the phase space from partial observations is to measure the dimension on time-delayed embeddings. While the theoretical effectiveness of time-delayed embedding is guaranteed by the Takens' theorem and variants [38, 35, 36], the method is empirically sensitive to noise and the choice of the embedding dimension, failing to deliver satisfactory results except for simple, low-dimensional systems. In this work, we aim at characterizing language models that are very high-dimensional and random in nature. For such systems, the time-delayed embeddings are often overwhelmed by observational noise and the dimension is overestimated.

Nevertheless, we observe that the correlation dimension values exhibit good consistency even if the time-delayed embedding is used. For the log-probability vector times series $x = [x_1, x_2, \cdots]$, we acquired two time-delayed sequences $x^{(3)}$ and $x^{(5)}$ with delays of 3 and 5, respectively. For a delay τ, $x_t^{(\tau)} = [x_t; \cdots; x_{t+\tau-1}]$, i.e., the the concatenation of vectors from t to $t+\tau-1$. The correlation integral curves for these time-delayed sequences are shown in Figure 8. While the time-delayed embeddings have 3x or 5x the dimension of the original sequence, the correlation integral curves are similar to that of the original sequence at small ε, except that the curves are shifted to the right. The slopes of the curves increased at large ε because of accumulated noise and is outside the range of interest.

This indicates that the next-token probabilities contain information beyond the next token, and it is sufficient to use the next-token probabilities to characterize the dimenison of the model's evolution.

D Supplementary Information for Experiments in Section 5

D.1 Degenerate Text Generation

The twenty prompts used in the experiment in Section 5.1 are listed in Table 5.

To generate different types of degenerate texts with GPT-4o, we gave different instructions to the model and collected its responses, as follows.

Table 4: Summary of the results on the SEP dataset for three pre-trained LLMs.

Article	**CorrDim**			**Perplexity**
	Qwen2.5-32B	Qwen2.5-7B	Falcon3-10B	Qwen2.5-32B
prisoner-dilemma	3.28	3.11	4.10	3.58
process-theism	4.42	3.56	8.49	5.68
cosmological-argument	4.93	5.26	5.30	5.83
information	5.22	5.28	5.41	6.31
ethics-chinese	5.39	5.07	6.18	5.52
recursive-functions	5.40	5.76	5.88	4.96
reasoning-automated	5.48	5.32	5.44	5.06
infinity	5.51	5.33	5.48	5.68
buddhism-tiantai	5.53	5.50	5.37	9.25
evil	5.64	6.01	6.37	4.69
logic-temporal	5.66	4.77	4.83	4.99
mohist-canons	5.72	6.06	6.14	3.75
linguistics	5.73	6.00	5.88	9.54
publichealth-ethics	5.79	5.96	5.77	8.87
mill-moral-political	5.79	6.12	6.06	4.38
gasset	5.85	6.33	6.55	11.22
computational-complexity	5.91	5.95	6.90	5.32
africana	5.91	5.32	7.05	12.21
principia-mathematica	5.93	5.39	6.17	5.36
levinas	5.96	6.14	6.73	15.92
formal-belief	6.03	5.99	7.19	5.82
spacetime-singularities	6.09	6.66	6.81	5.42
computational-linguistics	6.10	6.01	5.85	9.71
chaos	6.13	5.84	5.81	6.70
shaftesbury	6.17	5.93	6.14	8.62
possibilism-actualism	6.18	5.68	5.98	6.72
trinity	6.19	6.63	6.16	7.97
dynamic-epistemic	6.21	5.70	5.71	5.44
proof-theory	6.22	5.60	6.45	6.36
logics-for-games	6.30	5.52	5.53	9.92
game-theory	6.36	6.03	6.60	3.84
consciousness-temporal	6.38	6.13	6.83	8.63
sidgwick	6.40	6.55	6.78	10.32
epistemic-game	6.42	6.31	7.54	5.43
innateness-language	6.47	6.10	6.62	4.98
newton-philosophy	6.51	6.63	6.75	6.19
habermas	6.52	7.34	7.20	9.04
descartes-epistemology	6.62	6.56	6.32	6.48
russell-moral	6.65	6.62	6.98	7.04
mereology	6.67	6.96	6.71	7.71
connectives-logic	6.68	7.39	7.99	6.63
weyl	6.75	6.87	6.15	5.62
consciousness-intentionality	6.78	6.62	7.69	9.57
attention	6.79	6.56	6.65	7.10
margaret-cavendish	6.81	6.81	6.75	8.16
qm-action-distance	6.83	7.12	7.18	4.70
normativity-metaethics	6.84	7.06	6.01	5.63
epistemology-bayesian	6.84	6.94	6.95	5.94
aesthetics-18th-german	6.88	6.89	6.63	7.75
logic-inductive	6.97	6.76	7.16	5.10
chance-randomness	7.00	7.07	6.67	6.88
idealism	7.03	7.01	7.29	6.94
egalitarianism	7.07	6.20	7.46	3.14
reid	7.10	6.34	8.32	5.43
early-modern-india	7.15	6.35	6.74	4.70
heidegger-aesthetics	7.49	7.34	7.05	7.42
pythagoreanism	7.71	7.48	7.58	4.10
heidegger	7.84	8.69	7.59	6.59
al-farabi-metaphysics	8.33	9.31	8.70	6.04
ontological-commitment	8.33	8.10	7.25	5.19
Mean	6.31	6.27	6.56	6.71
Pearson's ρ (w.r.t. CorrDim of Qwen2.5-32B)	-	0.90	0.63	-0.01

Table 5: List of prompts used in the experiment of degeneration detection.

No.	Prompt
1	Describe the primary goals of an effective team.
2	Explain the basic steps involved in a standard project workflow.
3	Outline the advantages of using modern technology in daily life.
4	Discuss the key features of a reliable customer service program.
5	Summarize the benefits of maintaining a consistent work schedule.
6	Describe how a typical training session should be conducted.
7	Explain why clear communication is important in organizations.
8	Outline the main characteristics of a successful leadership style.
9	Discuss the factors that contribute to a smooth operational process.
10	Summarize the core principles of quality assurance.
11	Describe the role of feedback in performance improvement.
12	Explain the importance of setting realistic goals.
13	Outline the steps for conducting a standard evaluation.
14	Discuss how data is used to inform business decisions.
15	ummarize the benefits of a structured planning approach.
16	Describe the essential elements of a professional code of conduct.
17	"Explain how consistency can improve team efficiency.
18	Outline the main objectives of a routine maintenance program.
19	Discuss the value of transparency in reporting results.
20	Summarize the advantages of applying proven best practices.

Normal Responses We used the following instruction:

```
You are a creative text generator. Your task is to produce a
richly detailed, vivid, and engaging passage of at least 1,500
words (but no more than 2,000) that brings its subject to life
with specific imagery, sensory detail, unique examples, and
dynamic narrative. Avoid generic descriptions, stock phrases,
or ''safe'' language. Instead, strive for:
- Concrete specifics: name real or invented places, objects,
    characters, or processes.
- Sensory richness: evoke sight, sound, smell, taste, and touch
    wherever possible.
- Fresh metaphors and similes: craft original comparisons rather
    than clichés.
- Varied sentence rhythms: mix short, punchy lines with longer,
    flowing sentences.
- Emotional or intellectual hooks: give the reader something
    surprising, thought-provoking, or emotionally resonant.
Generate a single coherent text that feels alive and unmistakably
your own.''
```

Bland responses We used the following instruction:

```
You are a text generator. Produce a bland yet coherent passage
of 1,500-2,000 words that avoids exact repetition of phrases or words.
```

Incoherent responses We used the following instruction:

```
You are a text generator. Please produce a richly worded, vivid,
and non-repetitive passage (1,500-2,000 words) that is intentionally
disjointed-lacking logical coherence-so the reader feels the
narrative is fragmented.
```

Repetitive responses We used the responses generated under **Normal Responses** and then extracted the first sentence, and repeated it to form a passage of approximately the same length.

D.2 Random Text

For stress-testing the models, we used a string of random names, separated by commas, as the input text. The names are common English names, and we generated the string by sampling i.i.d. from the unique names. A part of the acquired random text is as follows:

```
Quinlan, Anthony, Henry, Felicity, Taylor, Raymond, Xander,
Christopher, King, Amanda, Flora, Nicole, Anthony, Frank,
Quiana, Owen, Finley, Paige, Victoria, Aaron, Ulrika, Sarah,
Ignacio, Emily, Yuna, Imogen, Cameron, Claire, William, Preston,
Ulrika, Sabrina, Neil, Zara, Joseph, Orion, Vivian, Quinn, Wyatt,
Paul, Sophia, Brian, Flynn, Hayden, Charles, Grace, Carter,
Heather, Quest, Jacob, Jordan, Frances, Griffin, Yasmin, Quiana,
Penelope, Emma, Sabrina, Elizabeth, Joseph, Zion, Quinlan, Omar,
Ruby, Virginia, Ursula, Flynn, Alexander, Ian, Griffin, Frances,
Yasmine, Warren, Isaiah, Ryan, Kyle, Xanthe, Lucy, Georgia, Gregory,
Ophelia, Georgia, Patricia, Xiomara, Kayla, Finley, Zayden, Noah,
Caitlin, Brittany, Connor, Quinton, Urban, King, Blake, Joshua,
...
```

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   Justification: Claims are supported by experimental results in Sections 3 through 5.

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   Answer: [Yes]

   Justification: The only hyperparameters used in the experiments are the range of $S(\varepsilon)$ to estimate the slope, which is described in Appendix A. All details of the dataset are explicitly described in the appendices.

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